

Print Sizing Guide - Portrait Orientation

Frame Size (in)	Recommended Mat Opening (in)	Paper Size (in)	Image Size (in)	Top & Side Borders (in)	Bottom Border (in)
8 x 10	4.5 x 6.5	8 x 10	4 x 6	2	2
11 x 14	7.5 x 9.5	11 x 14	7 x 9	2	3
16 x 20	10.5 x 13.5	16 x 20	10 x 13	3	4
18 x 24	12.5 x 18.5	18 x 24	12 x 18	3	4.5
24 x 36	19.5 x 29.5	24 x 36	19 x 29	4.5	5

Important Considerations

- **Mat Overlap:** The mat's opening is cut to be about 1/4 inch smaller on each side than the artwork it is designed to hold. This overlap ensures the print doesn't fall through the opening. Therefore, your image size should be slightly larger than the mat opening, but not so large that it is completely hidden.
- **Signature and Numbering Space:** The bottom border is always larger than the top and side borders. This is a traditional practice that helps to visually balance the artwork and provides ample space for your signature, edition number, and title. A good rule of thumb is to make the bottom border about 1.5 to 2 inches larger than the side and top borders.
- **Paper and Ink:** For limited edition prints, use high-quality, archival-grade paper and pigment-based inks (Giclée prints). This ensures the print's longevity and value.
- **Consistency:** For a true limited edition, every print in the run should be identical in size, paper, and color.
- **Centering:** When preparing your file for printing, ensure the image is centered horizontally on the paper. The vertical centering is what creates the larger bottom border.
- **Pixel Dimensions:** To ensure a high-quality print, your digital file should have a high resolution, typically **300 DPI** (dots per inch) at the final print size. For example, for an 8x10 print with a 4x6 image, your image file should be at least 1200x1800 pixels (4x300x6x300).

Print Sizing Guide - Landscape Orientation

This guide assumes your artwork is a horizontal rectangle. The frame sizes listed are in a standard landscape configuration (e.g., 10x8 inches).

Frame Size (in)	Recommended Mat Opening (in)	Paper Size (in)	Image Size (in)	Top & Side Borders (in)	Bottom Border (in)
10 x 8	6.5 x 4.5	10 x 8	6 x 4	2	2
14 x 11	9.5 x 7.5	14 x 11	9 x 7	2	3
20 x 16	13.5 x 10.5	20 x 16	13 x 10	3	4
24 x 18	18.5 x 12.5	24 x 18	18 x 12	3	4.5
36 x 24	29.5 x 19.5	36 x 24	29 x 19	4.5	5

Key Considerations for Landscape Prints

- **Mat Opening:** The mat opening for a landscape print is wider than it is tall, matching the orientation of your image. For instance, a mat for a 14x11 frame will have a 9.5x7.5 inch opening.
- **Signature Placement:** Just as with portrait-oriented prints, the bottom border is intentionally larger than the top and side borders. This provides the necessary space for you to sign, number, and title the print in pencil, which is a key part of its value as a limited edition.
- **Aesthetic Balance:** A larger bottom border, known as a '**plinth**', is a traditional and aesthetically pleasing choice for matted artwork. It provides visual weight and helps to draw the eye to the art itself.
- **Digital File Setup:** When setting up your digital file for printing, ensure the image is centered horizontally on the canvas. The extra space for the bottom border is accounted for in the vertical placement, where the image is placed closer to the top of the paper.

AI - Calculate the ideal print sizing guide based on the following information and constraints:

1. Image resolution of width x height px
2. Highest possible DPI over 300
3. Maintain aspect ratio
4. No cropping
5. Largest possible standard frame size with standard mat size
6. Larger bottom border for limited edition info (ie. series, title, signature)

The Fundamental Issue and Its Solution:

The core problem is that your image has a 3:2 aspect ratio (1.5), but standard mat openings (like 7.5x9.5 or 10.5x13.5) for common frames like 11x14 or 16x20 have a different aspect ratio (closer to 1.27-1.28). This makes it very difficult to fit your image *without cropping* while using a *standard mat* and keeping the image large.

The only way to strictly meet "No Cropping" and "Maintain Aspect Ratio" with a standard mat opening that doesn't match your image's aspect ratio is to:

1. Calculate the image's dimensions that *fit entirely within* the standard mat opening (meaning the mat opening will be larger than the image on one axis).
2. Ensure that the resulting image's DPI is still at least 300.

Let's retry with the **11x14 frame** and its standard **7.5" W x 9.5" H mat opening**, and make sure the image fits entirely inside.

- **Print Image Height (limited by mat opening height for no cropping):** Let's make the image's printed height 9.0 inches (to allow for 0.5" overlap with the 9.5" mat opening).
- **Print Image Width (maintaining aspect ratio):** $9.0 \text{ inches} / (3236/2157) \approx 9.0/1.5 = 6.0$ inches.
- **Check Mat Opening Width:** The mat opening is 7.5 inches wide. Your image is 6.0 inches wide. This works! The image fits, and the mat opening is wider than the image.
- **Check DPI:** Printing 3236 pixels at 9.0 inches high results in $3236/9.0 \approx 359.5$ DPI. This is excellent and well above your 300 DPI minimum.

This is the optimal solution for a standard 11x14 frame and standard mat.

5. Calculate Paper Size and Borders (Larger Bottom for LE Info):

- **Paper Size:** 11 x 14 inches (to fit the frame)
- **Image Dimensions:** 6.0 inches (width) x 9.0 inches (height)
- **Total Vertical Border Space:** $14 \text{ (paper height)} - 9.0 \text{ (image height)} = 5.0$ inches
- **Desired Bottom Border (for LE Info):** Let's target **2.75 inches** (a comfortable size for a 14-inch tall paper).
- **Remaining Top Border:** $5.0 \text{ (total vertical space)} - 2.75 \text{ (bottom border)} = 2.25$ inches.
- **Total Horizontal Border Space:** $11 \text{ (paper width)} - 6.0 \text{ (image width)} = 5.0$ inches
- **Side Borders (equal):** $5.0/2 = 2.5$ inches each.

Final Print Sizing Guide for the Attached Image

(Largest Practical Standard Frame & Standard Mat, Portrait)

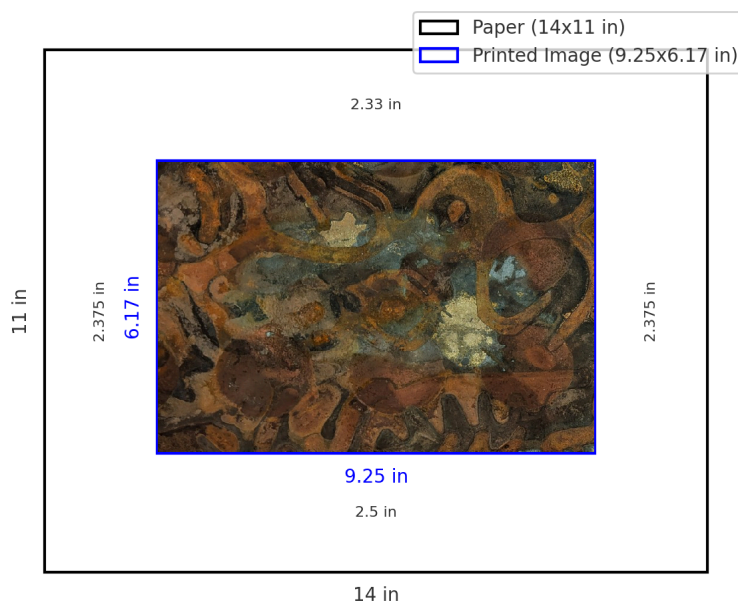
- **Image Resolution:** 2157 x 3236 pixels
- **Print Orientation:** Portrait
- **Target Standard Frame Size (Portrait):** 11 x 14 inches
- **Corresponding Standard Mat Outside Dimensions:** 11 x 14 inches
- **Corresponding Standard Mat Opening (typical, to fit an 8x10 photo):** 7.5 inches (width) x 9.5 inches (height)
 - *(This mat opening works perfectly: it is wider than your image, and its height accommodates your image with standard overlap, preventing any cropping.)*

Your Print's Specific Dimensions:

- **Print Image Width:** 6.0 inches
- **Print Image Height:** 9.0 inches
 - *(This ensures your entire image is visible, fits comfortably within the mat opening, and is printed at a high resolution of approximately 359.5 DPI, well above your 300 DPI minimum.)*
- **Recommended Paper Size (to fit frame):** 11 x 14 inches

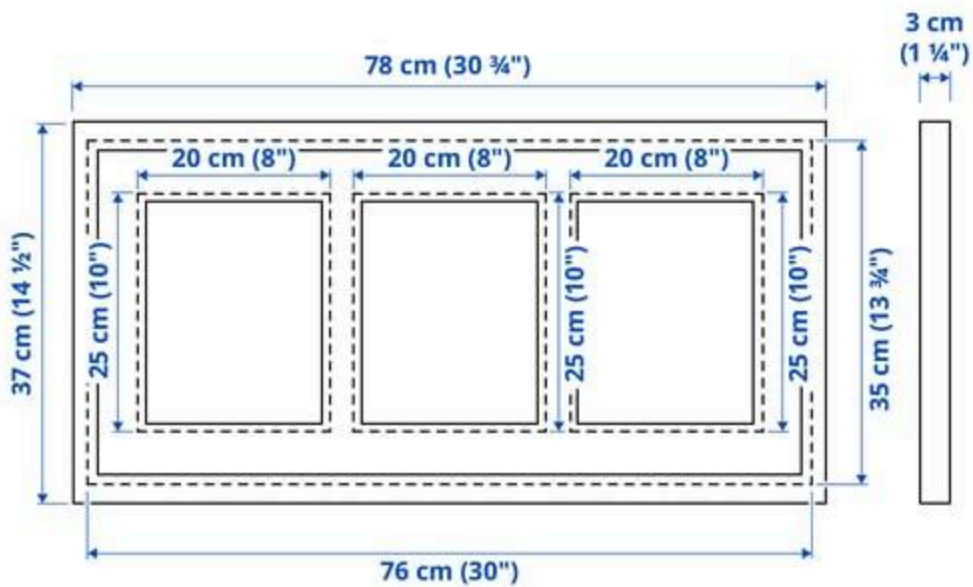
Paper Borders (from printed image edge to paper edge, for LE Info):

- **Side Borders:** 2.5 inches each (space between image and side edges of the paper)
- **Top Border:** 2.25 inches (space between image and top edge of the paper)
- **Bottom Border (for Signature/LE Info):** 2.75 inches (intentionally larger for limited edition details)



Measurements

- Picture without mat, width: 30 "
- Picture without mat, height: 13 $\frac{3}{4}$ "
- Picture with mat, width: 8 "
- Picture with mat, height: 10 "
- Mat opening, width: 7 $\frac{1}{2}$ "
- Mat opening, height: 9 $\frac{1}{2}$ "
- Frame width: 30 $\frac{3}{4}$ "
- Frame height: 14 $\frac{1}{2}$ "
- Frame, depth: 1 $\frac{1}{4}$ "



Q - Calculate the ideal print sizing guide based on the following information and constraints:

1. Image resolution of 4654 x 1659 pixels
2. Highest possible DPI above 250
3. Maintain aspect ratio
4. No cropping
5. Frame size of 30 x 13.75 in with mat opening of 26 x 9.5
6. Larger bottom border for limited edition info (i.e. series, title, signature)

Summarize the layout as follows:

- **Paper size:**
- **Printed image size:**
- **Side borders:**
- **Top border:**
- **Bottom border:**

A - Okay, here is the summarized layout based on the detailed calculations, directly addressing your request:

Print Sizing Layout

- **Paper Size:** 30" x 13.75"
- **Printed Image Size:** 18.62" x 6.64" (at 250 DPI)
- **Side Borders:** 5.69 inches (Left and Right, from frame edge to visible image edge)
- **Top Border:** 2.84 inches (From frame edge to visible image edge)
- **Bottom Border:** 4.27 inches (From frame edge to visible image edge - allowing for LE info)

Q - Generate an annotated line drawing of the print layout

A - Here's the updated line drawing for your **30" x 13.75" paper size:**

- **Paper size:** 30.00" x 13.75" (black rectangle)
- **Print size:** 18.62" x 6.64" (blue rectangle)
- **Side borders:** 5.69" each (centered horizontally)
- **Top border:** 2.84"
- **Bottom border:** 4.27" (nearly balanced, can be adjusted for LE info)

